

EXERCISES TO COMPLETE

Exercise 1 – MapPrinter

Download and complete the implementation of the `MapPrinter` Class.

```
import java.util.Map;
```

```
public class MapPrinter {
```

```
    public static void printKeys(Map<String, String> map) {
```

Check the `keySet()` method of the `Map` class and use it iterate over the keys and print them

```
    }
```

```
    public static void printValues(Map<String, String> map) {
```

Check the `values()` method of the `Map` class and use it iterate over the values and print them

```
    }
```

```
    public static void printPairs(Map<String, String> map) {
```

```
        for(Map.Entry<String, String> pair : map.entrySet()) {
```

Check the `getKey()` and `getValue()` methods of the `Map.Entry` class and use them to iterate over the keys and values of a `Map.Entry` object and print them

```
        }
```

```
    }
```

```
}
```

- Write a class `MapPrinterMain` to test your implementation.
- Create an instance `map` object and populate it with contact telephone numbers for your phone contacts list.

Hints:

- `keySet()` method of the `Map` Class:
<https://docs.oracle.com/javase/8/docs/api/java/util/Map.html#keySet-->
- `values()` method of the `Map` class:
<https://docs.oracle.com/javase/8/docs/api/java/util/Map.html#values-->
- `Map.Entry` interface:
<https://docs.oracle.com/javase/8/docs/api/java/util/Map.Entry.html>
- `HashMap` class in the Java Class Library
(<https://docs.oracle.com/javase/8/docs/api/java/util/HashMap.html>)

CMP5332 Lab 5 -Sheet (Arrays and Collections)

Exercise 2 - Scrabble Scorer

In the game of [Scrabble](#), players use tiles with letters to form words. Each letter is worth a certain number of points, and the score of a word is the total score of the letters. The points per individual letter are given in the table below:

A	B	C	D	E	F	G	H	I	J	K	L	M
1	3	3	2	1	4	2	4	1	8	5	1	3
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	1	3	10	1	1	1	1	4	4	8	4	10

Some tiles are blank; a blank tile can be used for any letter, but is worth 0 points. A blank tile will be represented using the underscore character, '_'.

Download and complete the implementation of the class below:

```
import java.util.*;

/*
 * This class scores words in the game of Scrabble.
 * A word's score is the total of its individual tile scores.
 */
public class ScrabbleScorer {
    private final Map<Character, Integer> tileScores;

    public ScrabbleScorer() {
        char[] tiles = "ABCDEFGHIJKLMNOPQRSTUVWXYZ".toCharArray();

        // TODO: initialise the array of individual letter scores
        int[] scores;

        // TODO: create the `tileScores` map
        // TODO: populate the `tileScores` map using the `tiles`
        // and `scores` arrays. Use a for loop that iterates over both arrays to create
        // the HashMap.
    }

    /*
     * Calculates the score for an individual letter tile.
     * If the character is not a letter, then 0 is returned.
     */
    public int scoreForTile(char tile) {
        // TODO: convert the tile to an uppercase character. Use the toUpperCase(char) method of
        // the Character class
        // TODO: check if the `tileScores` map contains this tile
        // TODO: get the tile score from the `tileScores` map
    }

    /*
     * Calculates the score for a word.
     */
    public int scoreForWord(String word) {
        char[] tiles = word.toCharArray();

        // TODO: calculate the total score of the tiles
    }

    /*
     * Returns the highest-scoring word from a list of words,
     * or null if the list is empty.
     */
    public String highestScoringWord(List<String> words) {
        // TODO: find the word with the highest score
    }
}
```

Hint: complete the TODO comments in the order they appear in the code ScrabbleScorerTest.

Additional Challenge - Test it!

Write a unit test class named ScrabbleScorerTest. You should test your scoreForTile, scoreForWord and highestScoringWord methods.

(Your submission will be considered complete even if the additional challenge is not submitted)

Exercise 3 - Scrabble Board Printer

You are required to write a class which prints out a Scrabble board. The board is represented as a 2D array of characters. Your class should allow Scrabble boards of any size, though you may assume the board is square.

For example, the following array represents a 5-by-5 Scrabble board, which you can use as a test case:

```
new char[][] {
    { '.', '.', 'W', '.', '.' },
    { '.', '.', 'O', '.', '.' },
    { '.', '.', 'R', '.', '.' },
    { 'H', 'E', 'L', 'L', 'O' },
    { '.', '.', 'D', '.', '.' }
}
```

Your class must print the board in a very specific format, as in the example below:

```
+-----+
| . . W . . |
| . . O . . |
| . . R . . |
| H E L L O |
| . . D . . |
+-----+
```

The tiles should be printed with a single space between tiles in a row, with dots . indicating empty squares on the board. The board has a border formed using the symbols +, - and |, with a single space between the border and the first and last tiles in each row.

Complete the `ScrabbleBoardPrinter` class below:

```
public class ScrabbleBoardPrinter {
    public static void printBoard(char[][] board) {
        // TODO
    }
}
```